

ASTR 5140: Project 2

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ABSTRACT

We investigate whether stellar populations associated with the Sagittarius (Sgr) tidal stream can be identified in Sloan Digital Sky Survey (SDSS) imaging of a field along the trailing arm of the stream. Using Dartmouth stellar evolution models, we construct fiducial isochrones representing the two dominant stellar populations of the Sgr stream: an old, metal-poor population (13 Gyr, $[\text{Fe}/\text{H}] = -1.8$, $[\alpha/\text{Fe}] = +0.2$) and an intermediate-age, metal-rich population (5 Gyr, $[\text{Fe}/\text{H}] = -0.5$, $[\alpha/\text{Fe}] = 0.0$). SDSS photometry for a $1^\circ \times 1^\circ$ target field and multiple control fields matched in Galactic latitude was downloaded via SQL queries. After dereddening the photometry and applying quality cuts, we identify candidate Sgr stream stars using two independent methods: (1) χ^2 proximity to the isochrones in color-magnitude space and (2) overlap between stellar photometric error boxes and the isochrone curves. Both approaches reveal a statistically significant overdensity of stars in the target field relative to control fields, particularly in the main-sequence turnoff (MSTO) region. Furthermore, Hess diagram residuals show CMD structures aligned with the expected isochrone locations. These results provide evidence for a Sgr-like stellar population in this region of the SDSS footprint, supporting the feasibility of follow-up studies targeting proper motions or deeper imaging.

1. INTRODUCTION

Tidal stellar streams provide powerful probes of the Milky Way’s gravitational potential and the history of its evolution. Among these, the Sagittarius (Sgr) tidal stream is one of the most prominent and extensively studied, tracing the ongoing disruption of the Sagittarius dwarf spheroidal galaxy as it orbits the Milky Way. Observations of the stream have revealed multiple stellar populations and complex spatial structure, making it a valuable laboratory for studying galaxy evolution and relevant dynamics.

Identifying Sgr stream stars in wide-field photometric surveys is challenging because the stream is embedded within dense foreground and background populations of Milky Way disk and halo stars. In many regions of the sky, the stream is detected statistically rather than through individual stellar memberships, often using color-magnitude diagram (CMD) analysis and comparisons to stellar population models.

In this project, we examine whether evidence for Sgr stream stars can be recovered in a region of Sloan Digital Sky Survey (SDSS) imaging located along the trailing arm of the stream. The goal is not to establish definitive membership for individual stars, but rather to determine whether a statistically significant stellar population consistent with Sgr properties can be identified. Such a detection would support the feasibility of follow-up observations to measure proper motions.

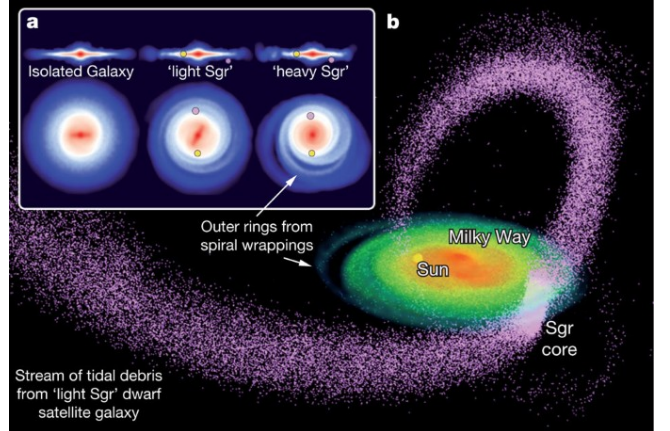


Figure 1. Diagram of the Sgr dwarf being stripped of its stars by the Milky Way (Purcell 2011).

2. BACKGROUND

The Milky Way did not form in isolation, but instead grew over billions of years by accreting and tidally disrupting smaller satellite galaxies. As these galaxies orbit the Milky Way, gravitational forces gradually pull stars away from them, forming elongated structures known as *tidal streams*. These streams trace the past orbital paths of the disrupted systems and provide a fossil record of the Galaxy’s assembly history. A visualization of the Milky Way and Sgr dwarf system is shown in Figure 1.

One of the most prominent examples is the Sagittarius tidal stream, which originates from the Sagittarius

dwarf spheroidal galaxy currently being torn apart by the Milky Way’s gravitational field. Over time, stars stripped from this galaxy have spread across large portions of the sky, forming leading and trailing arms that wrap around the Milky Way. Because these stars share a common origin, they exhibit similar distances, ages, and chemical compositions, allowing them to be studied as a coherent population.

Identifying stars belonging to the Sagittarius stream is challenging because they are mixed with far more numerous stars from the Milky Way’s disk and halo. In wide-field photometric surveys such as the SDSS, individual stream stars are generally not distinguishable based on position alone. Instead, astronomers rely on statistical methods that examine how groups of stars populate color–magnitude diagrams, which relate a star’s brightness to its color (a proxy for temperature and composition). Stars that formed together tend to follow predictable sequences in CMD space, described by stellar evolution models known as isochrones.

The goal of this project is to determine whether a stellar population consistent with the Sagittarius stream can be identified in SDSS imaging of a specific region of the sky. Establishing the presence of such a population is an important first step toward future follow-up observations, such as high-resolution imaging or proper-motion measurements, which could confirm membership and enable detailed studies of the stream’s structure and dynamics. Detecting the stream in this region would therefore contribute to broader efforts to map the Milky Way’s formation history and gravitational potential.

3. RELATED WORKS

Previous studies have mapped the Sgr stream using both photometric and spectroscopic surveys. [Belokurov et al. \(2006\)](#) and [Gibbons et al. \(2014\)](#) demonstrated that SDSS imaging can reveal stream structure through CMD overdensities and spatial correlations. Stellar population studies have shown that the Sgr stream contains multiple populations, including an old metal-poor component and an intermediate-age metal-rich component ([Siegel et al. 2007](#)). Chemical abundance measurements further constrain the α -element enrichment history of these populations ([Chou 2010](#)).

Isochrone fitting using stellar evolution models, such as the Dartmouth Stellar Evolution Database ([Dotter et al. 2008](#)), has become a standard approach for interpreting CMDs and identifying populations associated with tidal debris. Statistical techniques such as matched filtering and Hess diagram analysis have also been widely used to enhance the visibility of faint stream features against Milky Way contamination.

In the context of stream searching, works such as [Bonaca et al. \(2012\)](#) find that the presence of stream stars will materialize as an overdensity on the CMD. Thus, we aim to build on these approaches by combining CMD fitting, statistical overdensity measurements, and control-field comparisons to assess whether a Sgr-like population is detectable in a new SDSS field.

4. METHODS

4.1. SDSS Photometric Data

We downloaded photometric data from the SDSS DR19 PhotoPrimary catalog using SQL queries through the SkyServer API. The target field was defined as a $1^\circ \times 1^\circ$ region centered at (RA, Dec) = (02:09:37.4, -04:38:31.6). Thirteen control fields were selected at widely separated Galactic longitudes but matched in Galactic latitude to the target field to minimize differences in Milky Way foreground/background composition. The coordinates corresponding to each field are enumerated in Table 1.

Stars were selected for analysis using the following criteria:

1. SDSS object classification: stellar sources (type = 6)
2. clean photometry flags
3. magnitude limits: $g_0 < 23.5$, $r_0 < 23.5$
4. photometric uncertainty limits: $\sigma_g, \sigma_r < 0.15$

These limits were chosen to remain within the regime in which SDSS photometry maintains reliable uncertainties. At $g \gtrsim 23.5$, photometric errors increase rapidly and CMD structures become significantly broadened, reducing the reliability of isochrone comparisons. Initially, a SQL query was written to download only the target field, as described in Listing 1. Then, once the control fields were confirmed to match the target field in galactic latitude, a routine was written to perform multiple SQL queries via API request ¹ and download each control field as an individual .csv file ²³.

¹ Endpoint used: <https://skyserver.sdss.org/dr19/SkyServerWS/SearchTools/SqlSearch>

² Multi-field downloader and galactic coordinate validation available: https://github.com/zangston/ASTR-5140/blob/main/project2/sdss_data/download_sdss_fields.ipynb

³ Field files available: https://github.com/zangston/ASTR-5140/tree/main/project2/sdss_data/fields

Table 1. Target and control field centers in equatorial (J2000) and Galactic coordinates.

Field ID	α (deg)	δ (deg)	ℓ (deg)	b (deg)	Δb (deg)	$ \Delta b $ (deg)
glon_080	353.381110	-4.590060	80.000011	-60.791691	-2.644260×10^{-6}	2.644260×10^{-6}
glon_085	355.375477	-3.178909	85.000011	-60.791690	-1.899046×10^{-6}	1.899046×10^{-6}
glon_090	357.466454	-1.919136	90.000012	-60.791690	-1.828737×10^{-6}	1.828737×10^{-6}
glon_095	359.644412	-0.819254	95.000012	-60.791690	-2.174675×10^{-6}	2.174675×10^{-6}
glon_100	1.898647	0.113008	100.000010	-60.791689	-1.237323×10^{-6}	1.237323×10^{-6}
glon_105	4.217379	0.870862	105.000010	-60.791689	-1.359694×10^{-6}	1.359694×10^{-6}
glon_110	6.587807	1.448624	110.000011	-60.791689	-6.444923×10^{-7}	6.444923×10^{-7}
glon_115	8.996243	1.841851	115.000010	-60.791689	-4.922466×10^{-7}	4.922466×10^{-7}
glon_120	11.428305	2.047465	120.000009	-60.791689	-1.081309×10^{-6}	1.081309×10^{-6}
glon_125	13.869163	2.063840	125.000010	-60.791689	-8.724504×10^{-7}	8.724504×10^{-7}
glon_130	16.303813	1.890845	130.000010	-60.791689	-6.632705×10^{-7}	6.632705×10^{-7}
glon_135	18.717374	1.529851	135.000008	-60.791689	-6.196055×10^{-7}	6.196055×10^{-7}
glon_140	21.095369	0.983691	140.000008	-60.791688	-1.041506×10^{-8}	1.041506×10^{-8}
target	32.405833	-4.642111	166.038946	-60.791688	0	0

NOTE— $\Delta b \equiv b - b_{\text{target}}$, showing that all control fields are matched in Galactic latitude to within $\lesssim 3 \times 10^{-6}$ deg.

```

1 SELECT
2   p.objid,
3   p.ra, p.dec,
4   p.psfMag_u, p.psfMag_g, p.psfMag_r, p.
   psfMag_i, p.psfMag_z, -- Magnitudes
5   p.psfMagErr_u, p.psfMagErr_g, p.psfMagErr_r
   , p.psfMagErr_i, p.psfMagErr_z, -- Errors
6   p.extinction_u, p.extinction_g, p.
   extinction_r, p.extinction_i, p.
   extinction_z, -- Extinction
7   p.type, -- 6 = STAR
8   p.mode -- 1 = PRIMARY (unique object)
9 FROM PhotoPrimary AS p
10 WHERE
11   p.ra BETWEEN 31.905833 AND 32.905833
12   AND p.dec BETWEEN -5.142111 AND -4.142111
13   AND p.type = 6
14   AND p.mode = 1
15   AND p.clean = 1

```

Listing 1. Preliminary SQL query written to download the target field from the SDSS SkyServer

After the fields were downloaded, the aforementioned limits on photometric values and associated uncertainties were applied in Python after each file was loaded.

Finally, dereddened magnitudes were computed in Python using extinction values extracted via SQL:

$$g_0 = g - A_g, \quad r_0 = r - A_r \quad (1)$$

4.2. Isochrone Models

Fiducial isochrones were generated using the Dartmouth Stellar Evolution Database for two stellar populations associated with the Sgr stream:

1. 13 Gyr, $[\text{Fe}/\text{H}] = -1.8$, $[\alpha/\text{Fe}] = +0.2$

2. 5 Gyr, $[\text{Fe}/\text{H}] = -0.5$, $[\alpha/\text{Fe}] = 0.0$

Isochrones were produced in the SDSS ugriz photometric system and shifted into apparent magnitude space using a distance of 33.4 kpc, corresponding to a distance modulus: $\mu = 5 \log_{10}(d/10 \text{ pc}) \approx 17.62$

This is added to both g and r, so color is unchanged by the correction for distance.

4.3. Candidate Selection

Two independent methods were used to identify stars consistent with the Sgr stream populations. For either method, a subset of each field was taken that focuses on the region of the CMD corresponding to the main-sequence turnoff (MSTO) point of the isochrones, or $0.15 < (g - r)_0 < 0.65$ on the color axis and $20.0 < g_0 < 22.5$ on the magnitude axis. The MSTO region was emphasized because it provides the highest contrast between Sagittarius stream stars and Milky Way foreground populations. At brighter magnitudes, the CMD is dominated by foreground disk stars, while at fainter magnitudes, photometric uncertainties broaden, reducing discriminating power.

We note that CMD-based classification alone cannot uniquely identify Sagittarius stream membership for individual stars due to degeneracies between distance, metallicity, and photometric uncertainty. Therefore, our analysis focuses on statistical overdensities rather than individual classifications.

4.3.1. χ^2 Isochrone Proximity

For each star, a χ^2 statistic was computed relative to nearby isochrone points in two dimensions (magnitudes in g and r bands):⁴

$$\chi^2 = \left(\frac{g_0 - g_{iso}}{\sigma_g} \right)^2 + \left(\frac{r_0 - r_{iso}}{\sigma_r} \right)^2 \quad (2)$$

Stars with a χ^2 below the rejection threshold of 6.18 (corresponding to 95% confidence for two degrees of freedom) were classified as candidate stream members. The number of candidate stars as a proportion of all stars in the target field was computed:

$$f_{cand} = \frac{N_{cand}}{N_{total}} \quad (3)$$

Then, the same process of identifying candidates and computing a candidate fraction was repeated for the control fields, and the target field's candidate fraction was compared to that of the control fields.

4.3.2. Error-Box Overlap

Following the same method of comparing candidate fractions between the target and control fields, we define a simpler candidacy criterion where a star is considered a candidate if at least one isochrone point falls inside its 2-dimensional uncertainty rectangle in $(g - r)_0$ and g_0 ⁵. The error bar in the magnitude axis is simply the measured uncertainty in the g band, and the error bar in the color axis is the quadrature sum of the r and g bands:

$$\sigma_{(g-r)} = \sqrt{\sigma_g^2 + \sigma_r^2} \quad (4)$$

4.4. Hess Diagram Analysis

In addition to quantitative analysis of CMD geometry, Hess diagrams were constructed for the target and stacked control fields by binning stellar densities in CMD space. Residual diagrams (target minus controls) and smoothed density maps were used to identify CMD structures aligned with the isochrone predictions.

5. RESULTS

Plotting the target field stars with isochrones overlaid, with candidate Sgr stream stars highlighted according to both candidate selection methods, results in two very

similar CMDs. The results of both methods are shown in Figures 2 and 3.

The comparison results of both candidate selection methods are listed in Table 2.

Both methods of overdensity detection suggest the same conclusion – that the target field is consistent with the presence of a population compatible with Sgr stream properties. Additionally, visual inspection of the fields shows a markedly greater density in the MSTO region for the target field compared to control fields for both candidate selection methods, shown in Figures 4 and 5.

Using χ^2 -based selection restricted to the MSTO region, the candidate fraction in the target field exceeds the control-field mean by $S = 2.42\sigma$ (Table 2). The error-box overlap method produces a stronger overdensity with $S = 3.43\sigma$, indicating that the target CMD contains a statistically significant excess of stars consistent with the Sagittarius isochrones. The overdensity is clearly seen when plotting candidate fractions for both methods in histogram form, shown in Figures 6 and 7.

Hess diagram residuals show CMD density enhancements that align with the predicted locations of the 5 Gyr and 13 Gyr isochrones. While the signal is broadened by photometric uncertainties and foreground contamination, the excess follows the expected morphology of a Sgr-like stellar population. The Hess diagram results are shown in Figures 8 and 9.

Control fields matched in Galactic latitude do not show comparable overdensities, supporting the interpretation that the target field contains an additional stellar component consistent with Sgr stream populations. These results are consistent across independent candidate-selection methods and remain robust under variations in control-field sampling, indicating that the observed overdensity is unlikely to arise from statistical fluctuations in the Milky Way foreground population.

6. CONCLUSION

We have used SDSS photometry and stellar population modeling to search for evidence of Sagittarius tidal stream stars in a region of the SDSS footprint. Multiple independent analyses, including χ^2 isochrone fitting, uncertainty-overlap candidate selection, and Hess diagram residuals, consistently indicate a statistical overdensity of stars in the target field relative to control fields.

The CMD morphology of this excess aligns with expectations for Sgr stellar populations, particularly in the MSTO region where photometric precision is highest. While contamination from Milky Way foreground

⁴ χ^2 method available: <https://github.com/zangston/ASTR-5140/blob/main/project2/chi-squared.ipynb>

⁵ Error box overlap method available: <https://github.com/zangston/ASTR-5140/blob/main/project2/error-bar-overlap.ipynb>

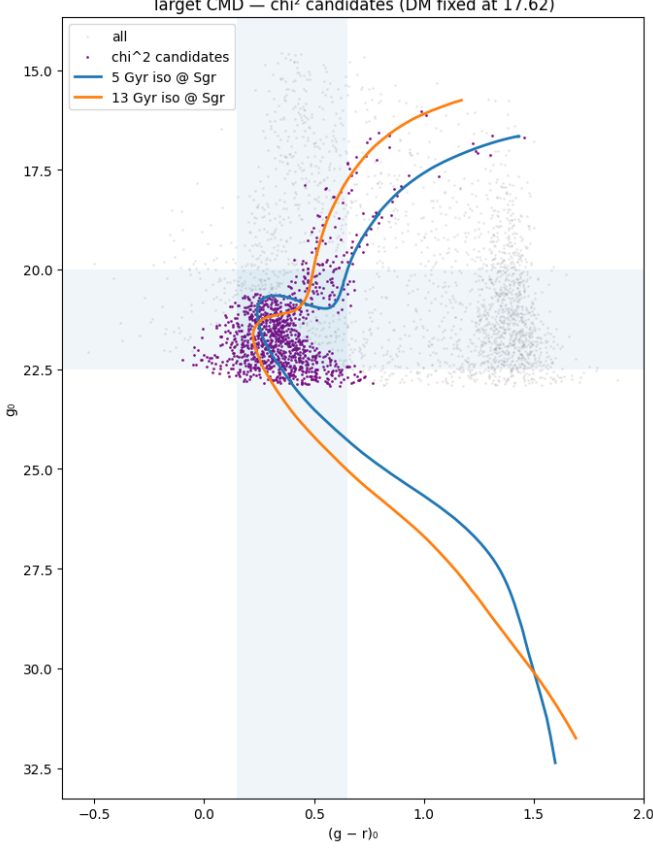


Figure 2. Target field CMD with candidate Sgr stream stars selected by the χ^2 method shown in purple; the MSTO selection region is lightly shaded.

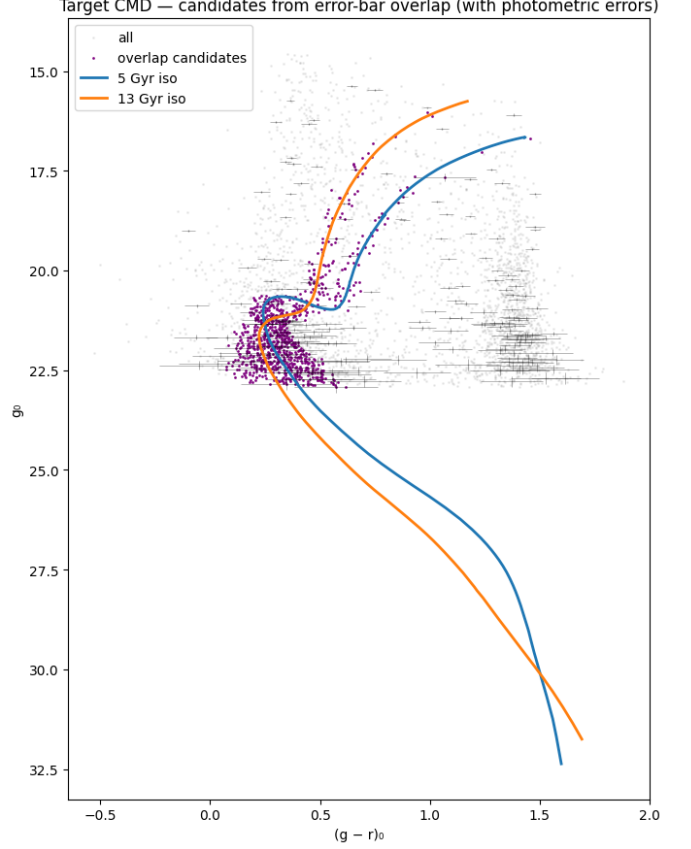


Figure 3. Target field CMD with candidate Sgr stream stars selected by the error-box overlap method shown in purple. A small subset of stars is drawn with error bars for visual reference.

Table 2. Candidate fractions in the MSTO region for the target field compared to the distribution across control fields.

Method	N_{MSTO}	N_{cand}	f_{target}	$\langle f_{\text{ctrl}} \rangle$	$\sigma(f_{\text{ctrl}})$	$S(\sigma)$
MSTO χ^2 candidates (fixed DM)	899	775	0.86207	0.76896	0.03845	2.42
MSTO overlap candidates	899	632	0.70300	0.57318	0.03781	3.43

NOTE— $f_{\text{target}} \equiv N_{\text{cand}}/N_{\text{MSTO}}$. Control-field summary statistics are computed over $n = 13$ control fields. The significance is defined as $S \equiv (f_{\text{target}} - \langle f_{\text{ctrl}} \rangle) / \sigma(f_{\text{ctrl}})$.

and halo stars remains significant, the consistency between multiple independent diagnostics strengthens the interpretation that the observed overdensity is not due to random CMD fluctuations but reflects an additional stellar population compatible with Sagittarius tidal debris.

These results do not establish definitive membership for individual stars but provide statistical evidence that the Sgr stream contributes to the stellar population in this region. This supports the feasibility of future follow-

up studies, such as proper-motion measurements using archival or new high-resolution imaging, to confirm stream membership and further characterize the stellar population.

Overall, this work demonstrates that CMD-based analysis of SDSS data can recover evidence for Sgr stream stars in previously unexplored fields at the population level, highlighting the continued value of wide-field photometric surveys.

7. FUTURE WORKS

Using CMD geometry alone, although seemingly effective, may still be too permissive an elucidator in characterizing stream stars. Future work could incorporate

spectroscopic metallicities and radial velocities to better distinguish Sagittarius stream stars from Milky Way populations. In addition, multi-epoch astrometric data such as those observed in [Sohn et al. \(2015\)](#) could provide proper-motion constraints that would allow individual stream membership to be confirmed.

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FIGURES

χ^2 -only candidates (fixed DM=17.62, $\chi^2 < 6.18$)

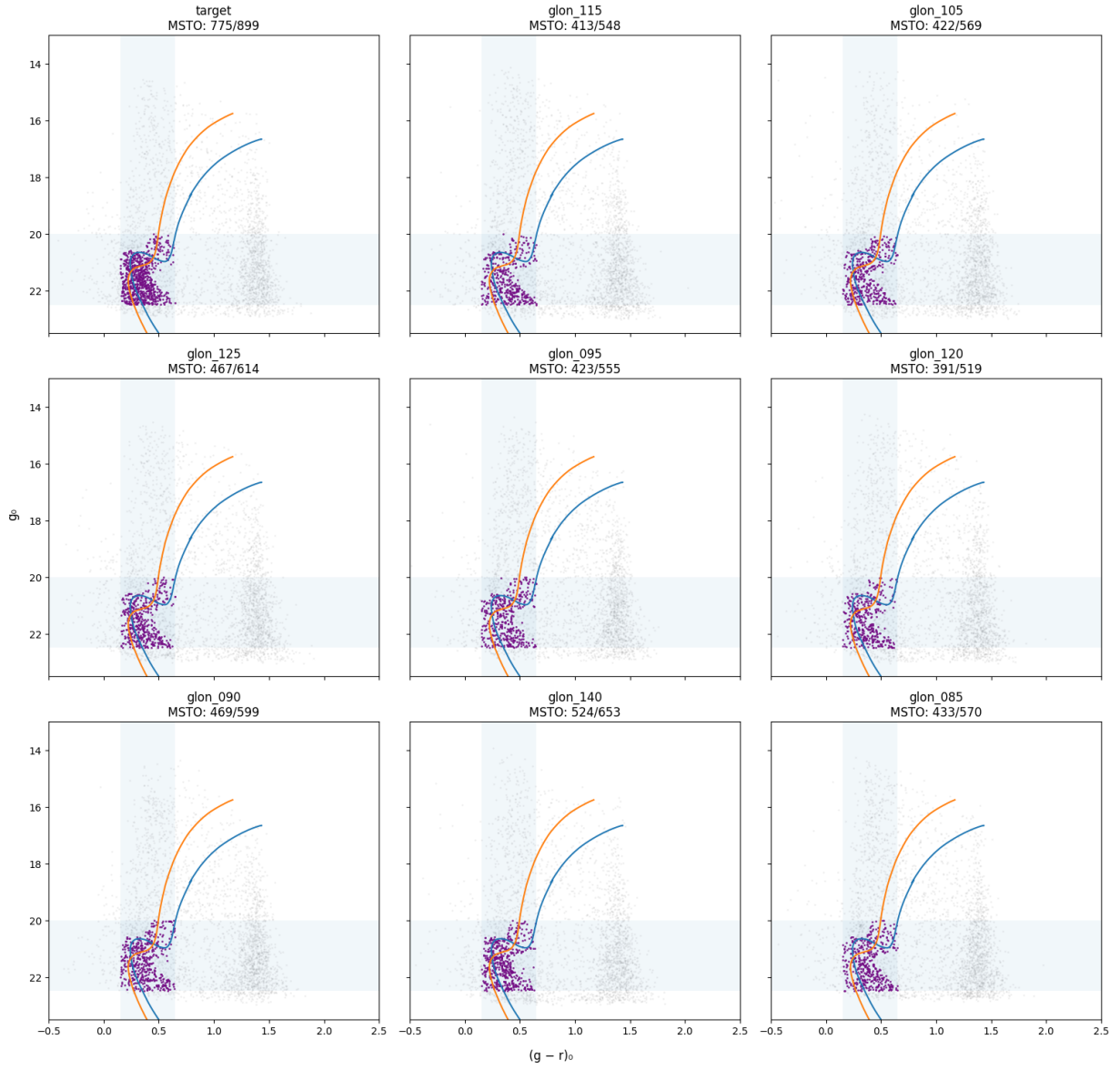


Figure 4. Target and control fields plotted with Sgr stream candidate stars according to χ^2 method.

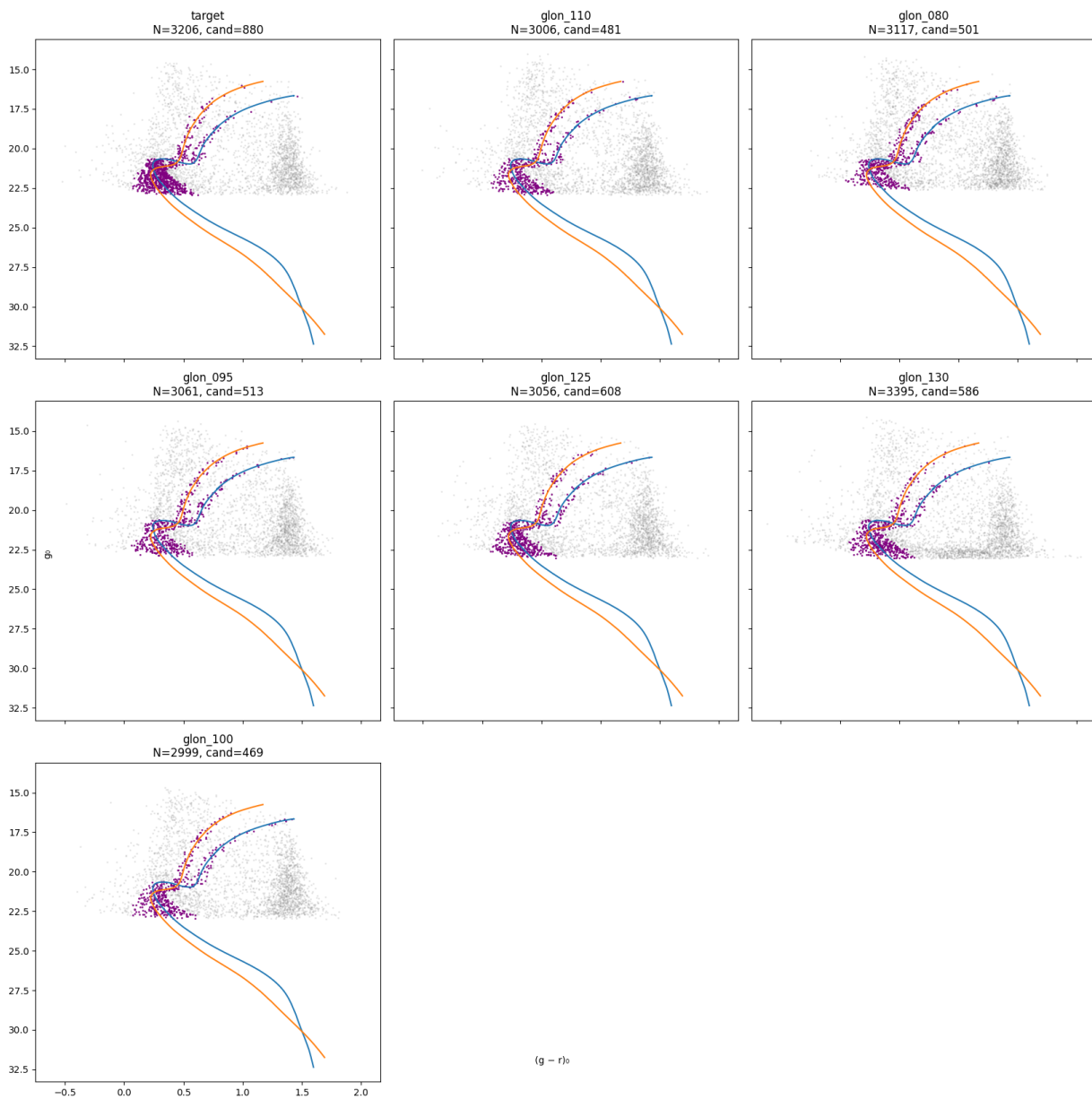


Figure 5. Target and control fields plotted with Sgr stream candidate stars according to error box method.

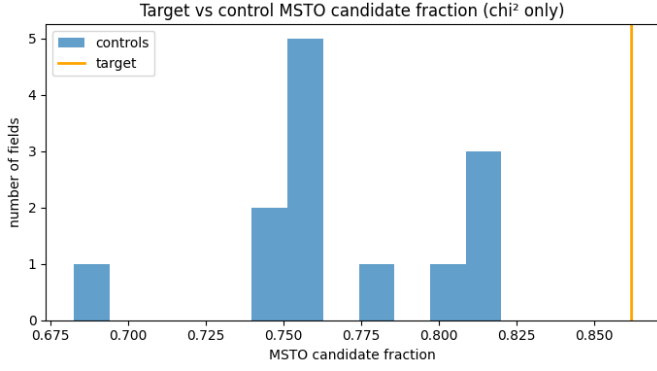


Figure 6. Histogram of MSTO candidate fractions across the 13 control fields using the χ^2 method. The orange line marks the target field candidate fraction.

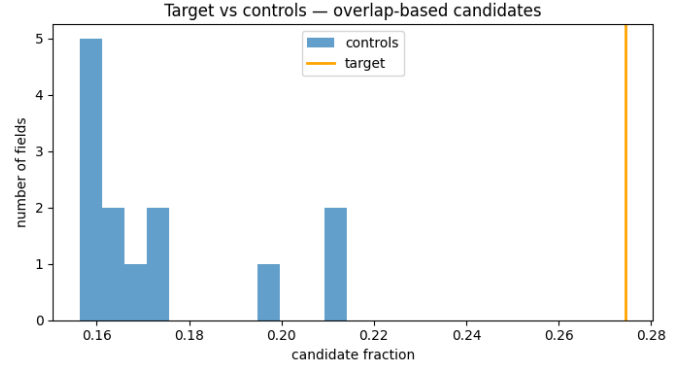


Figure 7. Histogram of MSTO candidate fractions across the 13 control fields using the error-box overlap method. The orange line marks the target field candidate fraction.

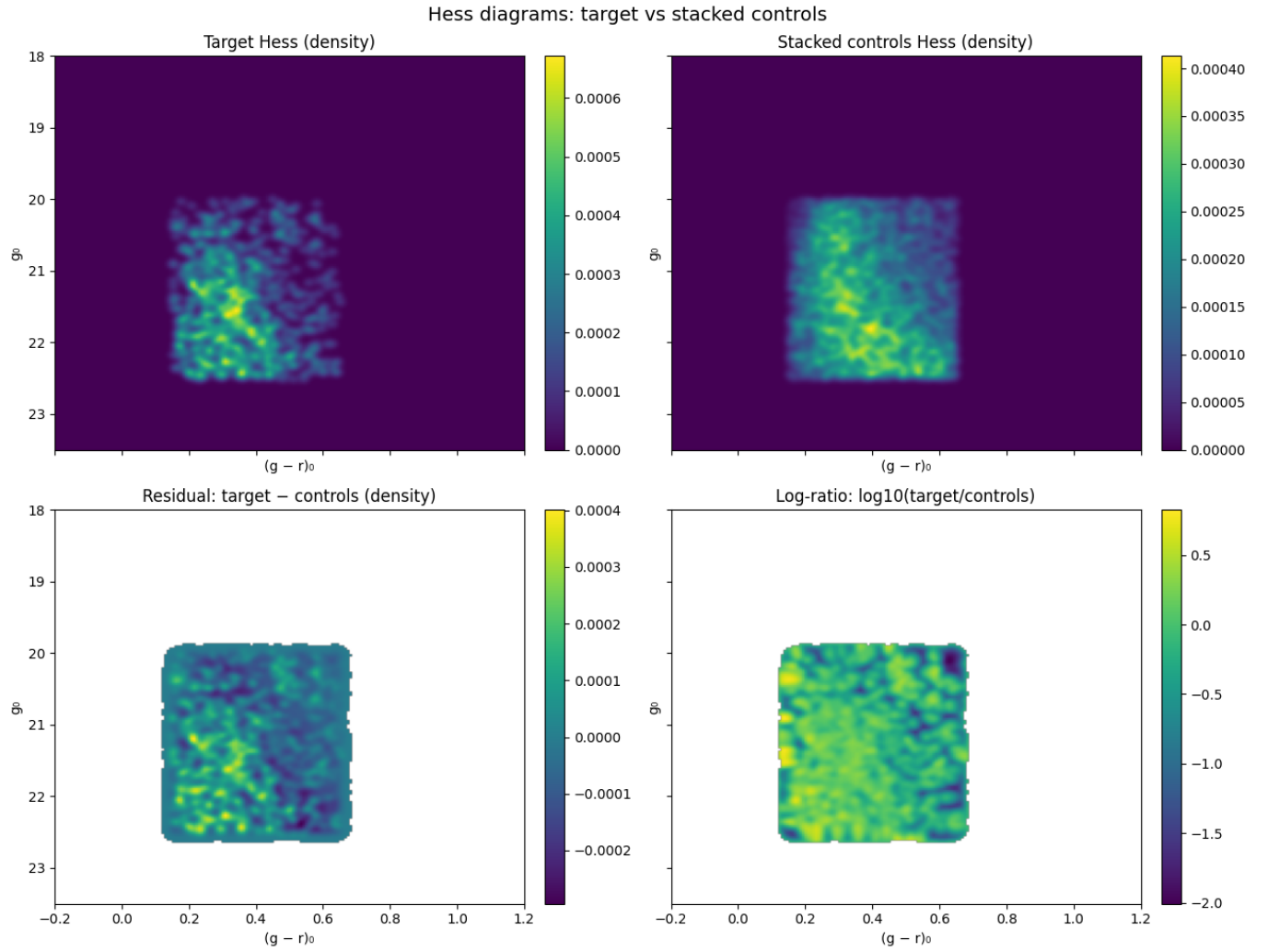


Figure 8. Hess diagrams for target field & stacked control fields, and resultant residuals.

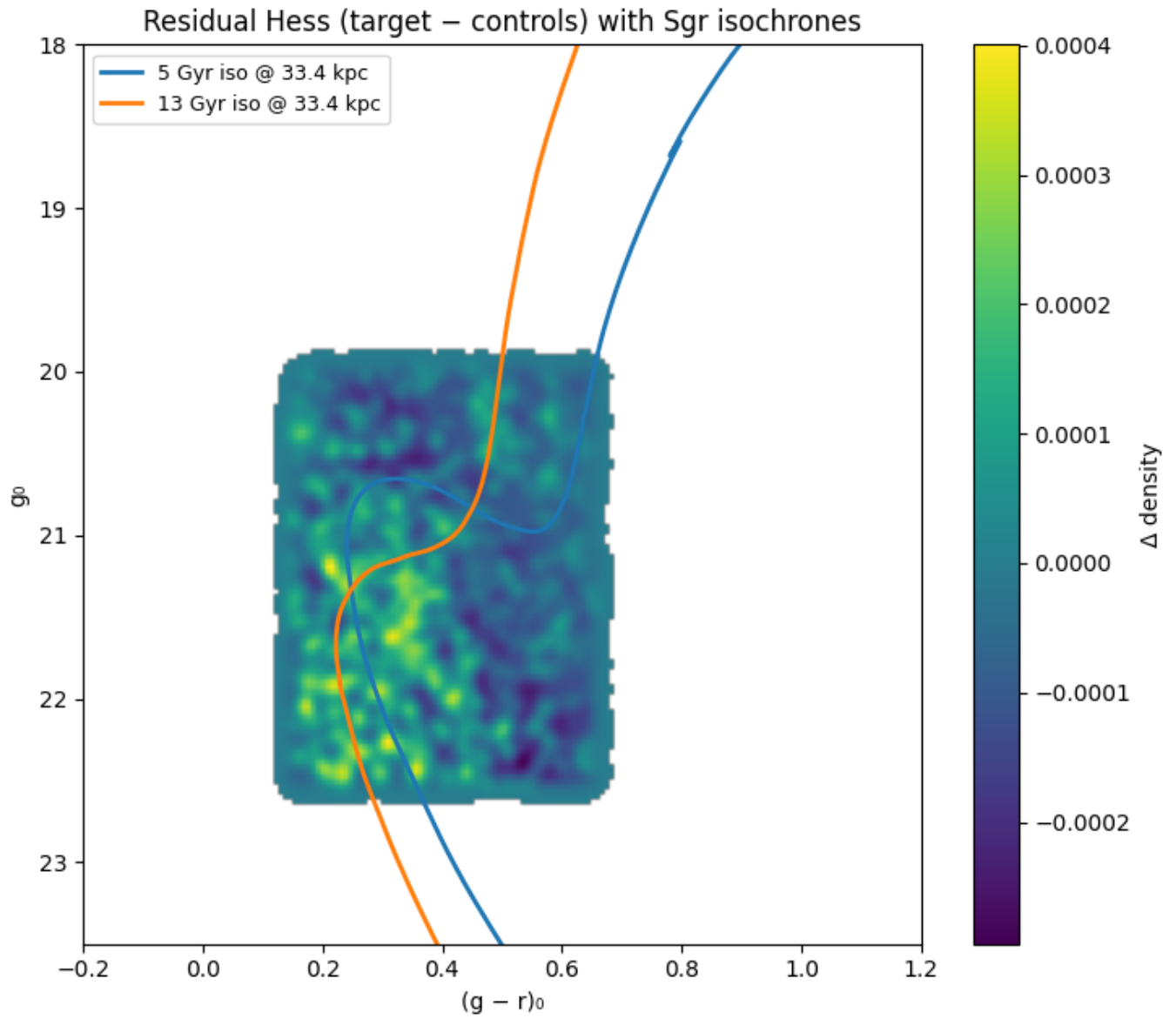


Figure 9. Gaussian-smoothed Hess diagram in MSTO region with Sgr stream isochrones superimposed.